

Fig. 43.1 Risk factors for chronic kidney disease (CKD). Nephrectomy performed for renal cell carcinoma is associated with nephron loss due to tissue removal, as well as ischemic and vascular injury. This loss of nephrons is associated with either new-onset CKD or an increased risk of progression of CKD in patients who have various risk factors, as noted. ESKD, End-stage kidney disease; GS, glomerulosclerosis; IF, interstitial fibrosis; HTN, hypertension; VS, vascular sclerosis.

Fig_43_1

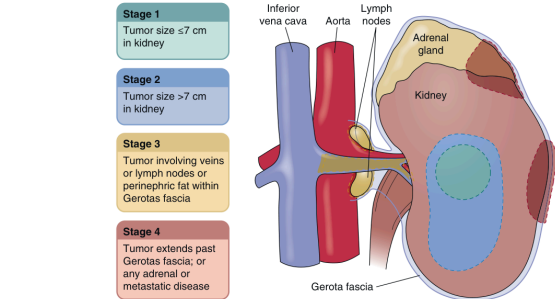


Fig. 43.2 The tumor-node-metastasis (TNM) staging classification. Shown is the general staging by the American Joint Committee on Cancer (AJCC) and Union for International Cancer Control (UICC), updated in 2017. (From Amin MB, Edge S, Greene F, et al., eds. *AJCC Cancer Staging Manual*. 8th ed. Springer International Publishing: American Joint Commission on Cancer; 2017 and Brierley JD, Gospodarowicz MK, Wittekind C, eds. *TNM Classification of Malignant Tumours*, 8th ed. Wiley-Blackwell; 2017.)

Fig_43_2

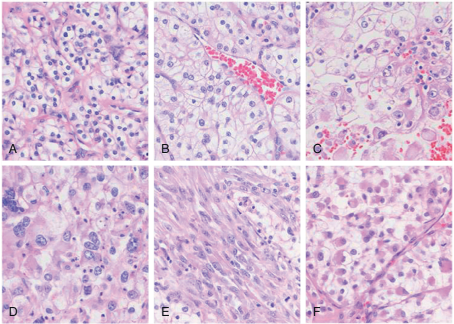


Fig. 43.3 The International Society of Urologic Pathology (ISUP) grading system for clear cell and papillary renal cell carcinoma (RCC). (A) Grade 1: Nucleoli are inconspicuous or absent. (B) Grade 2: Nucleoli are clearly visible at high-power magnification but are not prominent. (C) Grade 3: Nucleoli are prominent and are easily visualized at low-power magnification. (D) Grade 4: Presence of tumor giant cells and/or marked nuclear pleomorphism. (E) Grade 4: Sarcomatoid carcinoma. (F) Grade 4: Carcinoma showing rhabdoid differentiation.

Fig_43_3

Table 43.1 Bosniak Classification of Renal Cysts and Cystic Masses		
Classification	Description	Historical Likelihood of Malignancy (%) ¹⁴
Bosniak I	Simple cyst of water density with a thin wall that does not contain septa, calcifications, or other solid components	1.7
Bosniak II	Cyst may contain a few hairline septa and fine calcifications; uniform high-attenuation lesions that do not show contrast enhancement	18.5
Bosniak IIF	Cysts may contain multiple hairline thin septa or minimal smooth wall thickening	>18.5
Bosniak III	Thickened or irregular walls or septa with measurable enhancement	33
Bosniak IV	One or more enhancing nodules (>4 mm)	92.5

Table_43_1

Table 43.2 Subtypes of Renal Cell Tumors: World Health Organization Classification 2022	
Subtype	
Clear cell RCC	
Papillary RCC	
Oncocytoma	
Chromophobe RCC	
Collecting duct carcinoma	
Medullary carcinoma, not otherwise specified	
Acquired cystic disease-associated RCC	
Eosinophilic solid and cystic RCC	
Molecularly defined renal carcinomas:	
TFEB-rearranged RCC	
TFEB-altered RCC	
ELOC (formerly TCEB1)-mutated RCC	
Fumarate hydratase-deficient RCC	
Succinate dehydrogenase-deficient RCC	
ALK-rearranged RCC	
SMARCB1-deficient medullary-like RCC	
SMARCB1-deficient undifferentiated RCC, NOS	
SMARCB1-deficient dedifferentiated RCC of other specific subtypes	
RCC, Renal cell carcinoma.	

Table_43_2

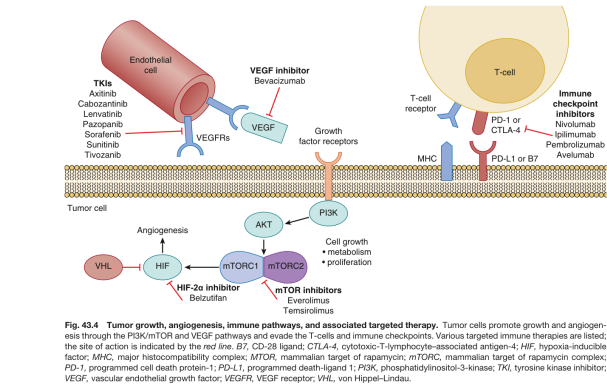


Fig. 43.4 Tumor growth, angiogenesis, immune pathways, and associated targeted therapy. Tumor cells promote growth and angiogenesis through the PI3K/mTOR and VEGF pathways and evade the T-cells and immune checkpoints. Various targeted immune therapies are listed; the site of action is indicated by the red line. B7, CD-28 ligand; CTLA-4, cytotoxic-T-lymphocyte-associated antigen-4; HIF, hypoxia-inducible factor; MHC, major histocompatibility complex; mTOR, mammalian target of rapamycin; mTORC, mammalian target of rapamycin complex; PD-1, programmed cell death protein-1; PD-L1, programmed death-ligand 1; PI3K, phosphatidylinositol-3-kinase; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor; VEGFR, VEGF receptor; VHL, von Hippel-Lindau.

Fig_43_4